

Shirin Shahabi

365-883-4747 | Toronto, ON | [Gmail](#) | [Personal Website](#) | [linkedin: shirin-shahabi](#) | [github: shirin-shahabi](#) | [Google Scholar](#)

PROFILE

Machine Learning Engineer with 5 years of hands-on experience in the Finance and Tech industries, holding an **MSc in Computer Science** from McMaster University. Proven track record in delivering production-grade AI solutions, enabling data-driven decision-making, and fostering cross-functional collaboration. Expert in **Reinforcement Learning, Verifiable AI Inference, Distributed Inference and Post-Training Optimization**. Specialized in AI Agent security, Distributed model Inference, and generative AI solution.

Cloud expertise in **AWS** (Lambda, EC2, SageMaker, Aurora, Airflow, Elasticsearch, EMR, Glue) with **Cloud Practitioner and ML associate certification** and **GCP** (Vertex AI, Kubernetes, VM instances).

EXPERIENCE

Machine Learning Engineer Inference Labs Inc

May 2024 – Present
Remote, Canada

- Contributed as a core developer to DSperse, an open-source framework for **distributed, verifiable inference, Improved inference time and optimized memory with fallback fault-tolerant mechanism** ([Sourcecode](#)).
- Collaborated on development and integration of JSTProve, an end-to-end pipeline for turning an ONNX model into a provable inference by applying quantization and compression, designed ONNX subgraph extraction, FFT decomposition, and advanced **graph manipulation (GraphSurgeon/Nvidia TensorRT)**([Sourcecode](#)).
- Developed the industry's first Verifiable AI Agent integrating proof of inference, Achieved a computational overhead reduction from 1000x to 8x via JIT access, Redis caching, and proprietary LLM evaluation metrics.
- Devised and implemented Prompt Injection detection and zk-proxy systems for **AI browser agents**, leveraging DSperse masking for secure LLM source verification; instead of Prompt Guardlines.
- Developed distributed, verified inference for an external **In-production, commercial Vision aviation model** using DSperse and JSTProve.
- Experimented with the first verifiable distributed LLM on Qwen 2.5, achieving near-zero computational overhead and automated testing via MCP, led to the creation of DSperse.
- Deployed Local DeepSeek-R1-671B parameters, quantized to 4-bit and pruned using structured and dynamic techniques with 17 tokens/s; Benchmarked against GCP-hosted equivalent using vLLM on a **Kubernetes** cluster.
- Engineered a production-ready SR&ED Retrieval-Augmented Generation (RAG) documentation pipeline utilizing Google Flan-T5, deployed on **Google VertexAI**.
- Implemented a Ranking Agent optimizing liquidity and slippage using DeepFM, and applying Direct Preference Optimization(**DPO**) and **GRPO fine-tuning**; containerized with **Docker** for scalable deployment.
- Updated a time-series LSTM model in production to mitigate model drift in TVL forecasting.

Machine Learning Developer McMaster University

Dec 2022 – Sep 2024
Hamilton, Canada

- Built an autonomous trading bot, with advanced risk mitigation strategies using Reinforcement Learning and FastDQN on TensorFlow; outperformed human traders by 185%.
- Designed and executed multi-objective network optimization for CSX Transportation Company using NetworkX and Gurobi; Scaled multi-threaded learning on HPC clusters (SLURM batch jobs).
- Engineered a novel **Hierarchical Reinforcement Learning framework**, improving conventional Multi-Armed Bandit solution by 13% with **Meta's PEARL**; a **Production-ready AI Agent** Library.
- Instructed "Statistical Data Analysis" course using Python at McMaster University.

Data Scientist Nobitex

Feb 2022 – Jan 2023
Remote

- Developed **scalable ensemble tree-based algorithms for customer clustering**, leveraging **PySpark** for distributed processing of 4+ million customers, doubling MAU from 4M to 8.5M within one year.
- Led cross-functional collaboration with the product team to design an **end-to-end marketing data modeling and ETL Pipeline** using Django and ORM Design.
- Defined over 50 financial KPIs, identifying company priorities and key metrics to support strategic growth.
- Developed blockchain-based fraud detection systems for OTLP wallet transactions and designed executive dashboards in Power BI.

- Utilized Web Engage for Customer Journey Mapping, achieving a 35% increase in conversion rate.
- Conducted market research and user survey analysis, driving a 10% increase in engagement.
- Performed sentiment analysis on Instagram and Twitter, extracting insights that improved retention rates by 30%.

Data Analyst

Oct 2020 – Dec 2021

Rocket Internet/SnappMarket

Remote, Canada

- Contributed to a \$100 million funded project for a **fully automated shopping experience** (Low-Cost Amazon Go), implementing indoor location tracking in collaboration with Rocket Internet.
- Developed an **optimized inventory reordering system** for 15 hypermarkets, automating algorithms for 8,000+ SKUs and coordinating supply chains across 200+ suppliers.
- Collaborated on **fraud detection**, retention, loyalty, and churn prediction programs through **AB Testing**.
- Mentored two interns within the Data Team, now pursuing Math PhDs at UWaterloo and UMichigan.
- Delivered over 50 optimized SQL queries benefiting Marketing, Operations, and Finance teams.
- Established real-time automated email reports ensuring data accuracy for strategic decision-making.

EDUCATION

M.Sc. Computer Science

Dec 2022 - Dec 2024

McMaster University

Thesis: Multi-objective Weight optimization through **Hierarchical Reinforcement Learning in Decision Making**.

- Member of Degroote finance and investment council ([DFIC Club](#)) - Quantitative Finance Department.
- Completed the [Vector Institute](#) 2024 Deep Learning and **Reinforcement Learning** Summer School.
- Master Thesis Presented at the Canadian Operational Research Society [CORS 2025](#) Conference.

B.Sc. Systems Engineering

Oct 2017 - Oct 2021

Sharif University

- Graduated with distinction from the Office of Exceptional Talents.

SKILLS

Core Expertise: Reinforcement Learning, Verifiable Inference, Distributed Inference, Quantization, RAG, Finetuning, AI Agents, MCP, Direct Preference Optimization, Prompt Injection, LLM Evaluation, Statistical Testing

Programming: Python (Pandas, Numpy, scikit-learn, XGBoost, networkx, Plotly), Rust, R (data.table, tidyverse, dplyr)

Data Engineering: SQL (MySQL, PostgreSQL), MongoDB, Redis, Vector Databases

Cloud Platforms: GCP (Kubernetes, Vertex AI), AWS Certified (Lambda, EC2, SageMaker, Airflow, Elasticsearch)

Deep/RL Learning: PyTorch, TensorFlow, Hugging Face Transformer, ONNX Runtime, OpenAI Gym, Ray, Pearl

LLM Inference: TensorRT, TensorRT-LLM, vLLM, NanoChat, MCP, DeepSpeed, AutoGen, DSpy, LlamaIndex

Data Tools/ DevOps: MLflow, Weights&Biases, PySpark, Docker, Kubernetes, Git, CI/CD, Rest API, Bash

PUBLICATION

JSTprove: Pioneering Verifiable AI for a Trustless Future ([Publication](#)) ([Sourcecode](#))

DSperse: A Framework for Targeted Verification in Zero-Knowledge Machine Learning ([Paper](#)) ([Sourcecode](#)).

HONORS AND AWARDS

CFA Institute Research Assistantship (\$10K)

Entrance Scholarship from Computational Science and Engineering (\$10K)

Excellence Scholarship Department of Mathematics at McMaster University(\$8K/year)

Merit-based Direct MBA-Finance Admission by the Office of Exceptional Talents at Sharif University

Member of Iran's National Elite Foundation (INEF)

Semifinalist in the National Mathematics Olympiad

National Youth Chess Champion